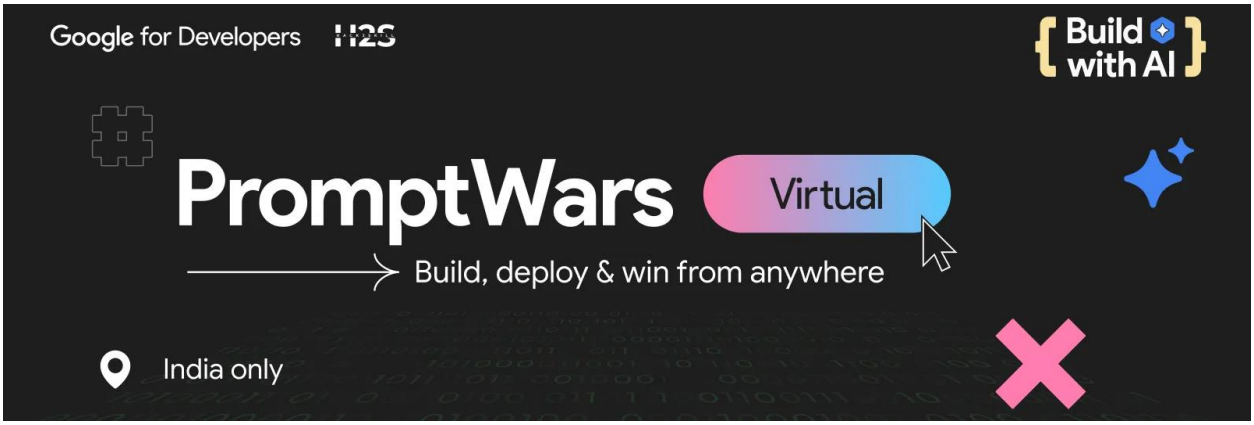


SmartCrowd AI: Intelligent Crowd Management System for Sporting Venues

📄 Abstract

SmartCrowd AI is a cloud-based system that improves crowd movement, reduces waiting time, and provides real-time updates in large venues. It uses Google Cloud Run and Docker for scalable and efficient performance.

❗ Problem Statement

Large venues face overcrowding, long queues, and poor coordination. There is a need for a smart system to manage crowd flow, reduce delays, and improve safety using real-time data.

⚙ Existing System

- Manual crowd control
- Static signboards
- No real-time updates
- High waiting time

🚀 Proposed System

- Real-time crowd tracking
- Smart navigation (low crowd routes)
- Queue prediction
- Cloud-based system using Google Cloud Platform

💻 Coding Part (Google Cloud)

🏟 SmartCrowd AI – Full Setup Guide

⚙ STEP 0: Prerequisites (VERY IMPORTANT)

Install these tools:

- Python (3.8 or above)
- Docker
- Google Cloud SDK

👉 Also create a project in

- Google Cloud Platform

📁 STEP 1: Create Project Folder

```
mkdir smartcrowd
cd smartcrowd
```

📄 STEP 2: Create Files

Create 3 files inside the folder:

🔹 app.py

```
from flask import Flask, jsonify, request
import random

app = Flask(__name__)

zones = {
    "Gate A": 50,
    "Food Court": 80,
    "Restroom": 30
}

@app.route("/")
def home():
    return "SmartCrowd AI Running!"

@app.route("/crowd")
def crowd():
```

```
    return jsonify(zones)

@app.route("/predict")
def predict():
    zone = request.args.get("zone")
    return jsonify({
        "zone": zone,
        "predicted_crowd": random.randint(10, 120)
    })

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=8080)
```

◈ requirements.txt

```
flask
gunicorn
```

◈ Dockerfile

```
FROM python:3.10-slim

WORKDIR /app
COPY . /app

RUN pip install -r requirements.txt

CMD exec gunicorn --bind :$PORT app:app
```

▶ STEP 3: Run Application Locally (Without Docker)

Install dependencies

```
pip install -r requirements.txt
```

Run app

```
python app.py
```

Open in browser

```
http://localhost:8080
```

Test APIs

```
http://localhost:8080/crowd
http://localhost:8080/predict?zone=Gate A
```

🐳 STEP 4: Build Docker Image

```
docker build -t smartcrowd .
```

▶ STEP 5: Run Docker Container

```
docker run -p 8080:8080 smartcrowd
```

👉 Open:

```
http://localhost:8080
```

☁ STEP 6: Login to Google Cloud

```
gcloud auth login
```

⚙ STEP 7: Set Project ID

```
gcloud config set project promptwarsproject-492814
```

🔌 STEP 8: Enable Required APIs

```
gcloud services enable run.googleapis.com
gcloud services enable cloudbuild.googleapis.com
```

🎁 STEP 9: Build & Push Image to Cloud

```
gcloud builds submit --tag gcr.io/promptwarsproject-492814/smartcrowd
```

🚀 STEP 10: Deploy to Cloud Run

```
gcloud run deploy smartcrowd \
--image gcr.io/promptwarsproject-492814/smartcrowd \
--platform managed \
--region asia-south1 \
--allow-unauthenticated
```

🌐 STEP 11: Get Live URL

After deployment, you will get:

https://smartcrowd-145365073250.asia-south1.run.app/

□ STEP 12: Test Live Application

Open in browser:

https://smartcrowd-145365073250.asia-south1.run.app/predict?zone=Food Court

https://smartcrowd-145365073250.asia-south1.run.app/crowd

Snapshot of Prototype

📸 Snapshot of Prototype

```
----------------------------
| SmartCrowd Dashboard     |
----------------------------
| Gate A     : High Crowd  |
| Food Court : Medium      |
| Exit Gate  : Low         |
----------------------------
| Suggested Route → Gate B |
----------------------------
```

8080-cs-1071122816006-default.cs-asia-southeast1-seal.cloudshell.dev

SmartCrowd AI Running!

console.cloud.google.com/welcome?project=promptwarsproject-492814&cloudshell=true

```
google.com%2Fauthcode.html&scope=openid+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.email+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcloud-platfo
rm+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fappengine.admin+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fsqlservice.login+https%3A%2F%2Fwww.googleapis.co
m%2Fauth%2Fcompute+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Faccounts.reauth&state=Q3fPXFtk4101E46HK9DyEewULKVozV&prompt=consent&token_usage=remote&acc
ess_type=offline&code_challenge=B8PQ8_5pBuY1Oi75Q7Xq7UmRvh5du0O7LHAgtRC4GXw&code_challenge_method=S256

Once finished, enter the verification code provided in your browser: 4/0Aci98E8hXfX42dlEULV4QG6ScyQ2jWpXe_TX27ECV2ax5NRiXFse4NJc0D4h7gAnki7U_Q

You are now logged in as [krithikgokuls@gmail.com].
Your current project is [promptwarsproject-492814].  You can change this setting by running:
  $ gcloud config set project PROJECT_ID
krithikgokuls@cloudshell:~/smartcrowd (promptwarsproject-492814)$ gcloud config set project promptwarsproject-492814
Updated property [core/project].
krithikgokuls@cloudshell:~/smartcrowd (promptwarsproject-492814)$ gcloud services enable run.googleapis.com
gcloud services enable cloudbuild.googleapis.com
```

smartcrowd-145365073250.asia-south1.run.app

SmartCrowd AI Running!

console.cloud.google.com/welcome?project=promptwarsproject-492814&cloudshell=true

```
60e70dddd9ea: Preparing
60e70dddd9ea: Layer already exists
39b3f7475ebd: Layer already exists
af8e2cb9023a: Layer already exists
e35db0322204: Layer already exists
ee31df5ce618: Pushed
f96342c327ef: Pushed
32339c107df9: Pushed
latest: digest: sha256:bd2450d722173e33962132e526212bae68f49c5a8d6e4b22aafc52b5d74468a2 size: 1784
DONE
---------------------------------------------------------------------------------------------------------------
ID: e5d853e1-1ad3-450b-b2bf-43699584464f
CREATE_TIME: 2026-04-09T15:05:40+00:00
DURATION: 39S
SOURCE: gs://promptwarsproject-492814_cloudbuild/source/1775747136.273773-7581f1339e794dc193b116b8bed06ca3.tgz
IMAGES: gcr.io/promptwarsproject-492814/smartcrowd (+1 more)
STATUS: SUCCESS
krithikgokuls@cloudshell:~/smartcrowd (promptwarsproject-492814)$
```

smartcrowd-145365073250.asia-south1.run.app/predict?zone=Food%20Court

Pretty-print

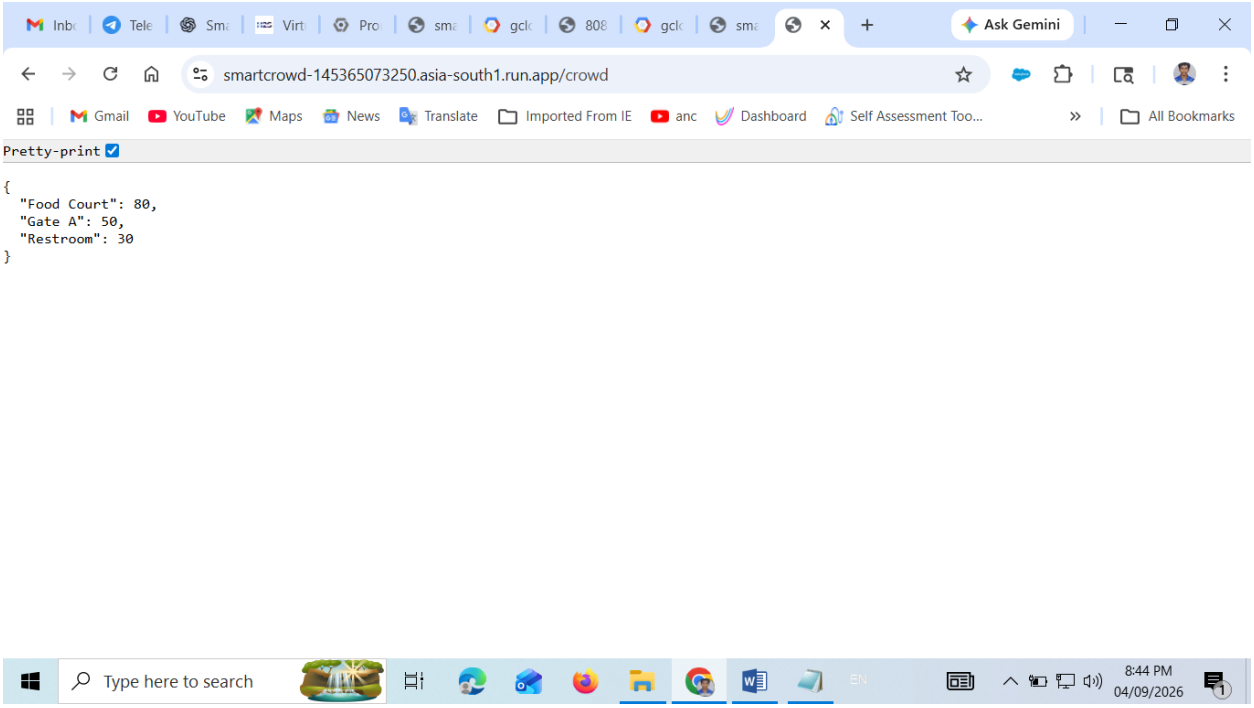

```
{
  "predicted_crowd": 73,
  "zone": "Food Court"
}
```


console.cloud.google.com/welcome?project=promptwarsproject-492814&cloudshell=true

Google Cloud | PromptWarsProject

CLOUD SHELL Editor

```
--image gcr.io/YOUR_PROJECT_ID/smartcrowd \
--platform managed \
--region asia-south1 \
--allow-unauthenticated^C
krithikgokuls@cloudshell:~/smartcrowd (promptwarsproject-492814)$ gcloud run deploy smartcrowd \
--image gcr.io/promptwarsproject-492814/smartcrowd \
--platform managed \
--region asia-south1 \
--allow-unauthenticated
Deploying container to Cloud Run service [smartcrowd] in project [promptwarsproject-492814] region [asia-south1]
OK Deploying new service... Done.
  OK Creating Revision...
  OK Routing traffic...
  OK Setting IAM Policy...
Done.
Service [smartcrowd] revision [smartcrowd-00001-tcq] has been deployed and is serving 100 percent of traffic.
Service URL: https://smartcrowd-145365073250.asia-south1.run.app
krithikgokuls@cloudshell:~/smartcrowd (promptwarsproject-492814)$
```



✅ Advantages

- Real-time monitoring
- Reduced waiting time
- Better safety
- Scalable cloud system

🌍 Applications

- Stadiums
- Concerts
- Airports
- Shopping malls

🔮 Future Scope

- AI-based prediction
- AR navigation
- Emergency alerts system

🏁 Conclusion

SmartCrowd AI improves event experience by reducing congestion and providing real-time insights using cloud technology.

📚 References

- Google Cloud Documentation
- Flask Documentation
- IoT Research Papers